Exact first separation in the Thue–Morse run-length sequence

Carptopus
carptopus@163.com

Manuscript, 2 September 2026

Internally verified candidate manuscript, version 0.1-beta. The mathematical claims have passed project-internal zero-trust proof and manuscript audits; external review is pending.

Abstract

Let $d = (d(n))_{n\geq 0}$ be the adjacent-run-length sequence of the Thue–Morse word, equivalently the fixed point beginning in 1 of $1 \mapsto 121$, $2 \mapsto 12221$. For $q \geq 2$, we determine exactly the first position at which the two kernel subsequences $d(2^{q+2}n)$ and $d(2^q n)$ differ. Writing

$$E_q = \begin{cases} 2^{q+1} - 2, & q \text{ even}, \\ 2^{q+1} + 1, & q \text{ odd}, \end{cases}$$

we prove that the first position is $\lfloor 2^{E_q}/9 \rfloor$. The proof combines the canonical Jacobsthal representation with a scale-stability lemma and an extremal alternating-binary-digit argument. We extend the result to arbitrary distances in the same 4-scale chain and derive double-exponential lower bounds when the odd part of the multiplier is fixed and its 2-adic valuation grows, with an exact infinite family governed by multiplicative orders.

1 Main results

Let

$$N_q = \min\{n \geq 1 : d(2^{q+2}n) \neq d(2^q n)\}.$$

Theorem 1. *For every $q \geq 2$,*

$$N_q = M_q := \left\lfloor \frac{2^{E_q}}{9} \right\rfloor, \qquad E_q = \begin{cases} 2^{q+1} - 2, & q \text{ even}, \\ 2^{q+1} + 1, & q \text{ odd}. \end{cases} \tag{1}$$

The cases $q = 3, 4$, giving 14563 and 119304647, were recorded by Allouche, Allouche and Shallit. The contribution of Theorem 1 is the exact all-q formula and its minimality proof, not the discovery of those individual values or the eventual separation of kernel subsequences.

For $r \geq 1$, define in the extended positive integers, with $\inf \varnothing = +\infty$,

$$R_r(b) = \inf\{n \geq 1 : d(4^{r+1}bn) \neq d(4bn)\}.$$

Theorem 2. *Write $b = 2^j u$, where u is odd, and set $q = j + 2$. Then for every $r \geq 1$,*

$$R_r(2^j u) \geq \left\lceil \frac{M_q}{u} \right\rceil. \tag{2}$$

If $u \mid M_q$, equality holds. Moreover, a fixed odd u divides M_q for infinitely many even q if and only if

$$\nu_2(\mathrm{ord}_{9u}(2)) \leq 1. \tag{3}$$

2 Two encodings of the sequence

Let

$$V = \{v \geq 1 : \nu_2(v) \equiv 0 \pmod 2\}, \qquad F(x) = |V \cap [1, x]|,$$

and define

$$e(x) = 3F(x) - 2x, \qquad H(x) = 2x - F(x) = \frac{4x - e(x)}{3}.$$

Then H is strictly increasing and

$$e(2x) = -e(x), \qquad e(2x+1) = 1 - e(x). \tag{4}$$

The Thue–Morse word changes between positions $n-1$ and n exactly when $\nu_2(n)$ is even. Hence, if $v_1 < v_2 < \cdots$ lists V, then $d(k) = v_{k+1} - v_k$. A direct counting argument shows

$$d(k) = 2 \quad \Longleftrightarrow \quad k \in H(V). \tag{5}$$

If $x = \sum_j \epsilon_j 2^j$, recursion (4) gives

$$e(x) = \sum_j (-1)^j \epsilon_j. \tag{6}$$

This converts separation of two values of d into instability of membership in $H(V)$ under multiplication or division by four.

The equivalent Jacobsthal encoding uses $J_i = (2^i - (-1)^i)/3$. For a binary word $w = b_m \cdots b_1$, put $[w]_J = \sum b_i J_i$. Lemma 3.7 of Allouche–Allouche–Shallit gives a unique representation that starts in 1 and ends in an even (possibly zero) number of 1's; we call it canonical. Their Theorem 3.8 states, in our indexing convention, that $d(n) = 1$ exactly for $n = 0$ or when a Jacobsthal representation beginning in 1 ends in a positive even number of 0's or a positive even number of 1's; otherwise $d(n) = 2$. If $B(w) = \sum b_i 2^{i-1}$ and $S(w) = \sum b_i (-1)^i$, then

$$3[w]_J = 2B(w) - S(w), \qquad [w00]_J = 4[w]_J + S(w). \tag{7}$$

3 Explicit witnesses

When q is even, set

$$w_q = (10)^{2^q-2}1^q, \qquad z_q = (10)^{2^q}0^{q-2}.$$

When q is odd, set

$$w_q = (10)^{2^q}0^{q-1}, \qquad z_q = (10)^{2^q-1}0100\,1^{q-1}.$$

Each displayed word is canonical under Lemma 3.7: for even q, w_q ends in q ones and z_q ends in zero ones; for odd q, w_q ends in zero ones and z_q ends in $q-1$ ones. Summing the geometric progressions in (7) gives

$$[w_q]_J = 2^q M_q, \qquad [z_q]_J = 2^{q+2}M_q.$$

For even q, the terminal constant blocks have lengths q and $q-1$; for odd q, they again have lengths q and $q-1$, in the opposite symbols. Thus Theorem 3.8 gives

$$d(2^q M_q) \neq d(2^{q+2}M_q).$$

Thus $N_q \leq M_q$.

4 Scale stability

Lemma 3. *Let $v \in V$.*

1. *If $2^r \mid H(v)$, $r \geq 2$, and $|e(v)| < 2^r$, then $4H(v) \in H(V)$.*
2. *If $2^r \mid H(v)$, $r \geq 4$, and $|e(v)| < 2^{r-2}$, then $H(v)/4 \in H(V)$.*

Proof. Write $e = e(v)$. Since $3H(v) = 4v - e$ and $2^r \mid H(v)$, we have $4 \mid e$. First suppose $e > 0$. Write

$$v = 3 \cdot 2^{r-2}a + e/4.$$

Because $0 \leq e/4 < 2^{r-2}$, this is a carry-free high/low binary decomposition. Put

$$T = 3 \cdot 2^r a + e/4 = 4v - 3e/4.$$

Thus T is obtained from v by inserting two zeroes between the two blocks. Equation (6) and the unchanged trailing-zero count give $e(T) = e(v)$ and $T \in V$, while

$$H(T) = \frac{4T - e(T)}{3} = 4H(v).$$

If $e = -4b < 0$, set $k = r - 2$ and write

$$v = 2^k(3a-1) + (2^k - b), \qquad T = 2^{k+2}(3a-1) + (2^{k+2} - b).$$

For $0 < b < 2^k$, the identity

$$e(2^k - b) = e(2^k - 1) - e(b - 1)$$

shows that the two low blocks have the same alternating digit sum because k and $k+2$ have the same parity; both also have 2-adic valuation $\nu_2(b)$. Hence again $e(T) = e(v)$, $T \in V$, and $H(T) = 4H(v)$. For $e = 0$, take $T = 4v$. This proves the first assertion.

For the second assertion, $|e|/4 < 2^{r-4}$ leaves two genuine separating zeroes that can be deleted. Precisely, put

$$R = (v + 3e/4)/4.$$

If $e = 4b > 0$, then

$$v = 3 \cdot 2^{r-2}a + b, \qquad R = 3 \cdot 2^{r-4}a + b,$$

with $0 < b < 2^{r-4}$. If $e = -4b < 0$, then

$$v = 2^{r-2}(3a - 1) + (2^{r-2} - b), \qquad R = 2^{r-4}(3a - 1) + (2^{r-4} - b).$$

The same identity, now with exponents $r-2$ and $r-4$, gives $e(R) = e(v)$ and $\nu_2(R) \equiv \nu_2(v) \pmod 2$. For $e = 0$, simply take $R = v/4$. Thus $R \in V$ in every case and $H(v) = 4H(R)$, proving the second assertion. $\square$

It follows that if $2^q \mid x$ and $d(4x) \neq d(x)$, then the member side $x = H(v)$ or $4x = H(v)$ must satisfy

$$|e(v)| \geq 2^q. \tag{8}$$

5 The extremal alternating-digit argument

Let

$$U_m = (10)^{m-1}1 = \frac{4^m - 1}{3}.$$

Then $e(U_m) = m$. Among L binary positions there are only $\lceil L/2 \rceil$ positive-sign positions and $\lfloor L/2 \rfloor$ negative-sign positions in (6); therefore every positive integer with $|e(x)| \geq m$ needs at least $2m - 1$ binary digits. Equality with positive sign is attained uniquely by U_m.

Set $m = 2^q$ and

$$v_q = 2^{s_q} U_m, \qquad s_q = \begin{cases} q - 2, & q \text{ even}, \\ q - 1, & q \text{ odd}. \end{cases} \tag{9}$$

The congruence $e(v) \equiv 4v \pmod{2^r}$, together with the digit-length bound, proves the following extremal statement.

Lemma 4. *If $v \in V$, $|e(v)| \geq 2^q$, and $2^q \mid H(v)$, then $v \geq v_q$. Under the stronger divisibility $2^{q+2} \mid H(v)$, the same minimum holds for even q; for odd q, every candidate on that member side satisfies $H(v)/4 > H(v_q)$.*

Proof. We use

$$e(v) \equiv 4v \pmod{2^r}. \tag{10}$$

First assume $2^q \mid H(v)$ and, for a contradiction, $v < v_q$. The digit-length bound reduces the possibilities to

$$e(v) = m + 4a \quad \text{or} \quad e(v) = -m - 4a, \tag{11}$$

where $a \geq 0$. For $q \geq 3$, the available number of positive or negative digit positions gives $4a \leq (q+1)/2 < 2^q$, and hence $a < 2^{q-2}$; when $q = 2$, only $a = 0$ remains. Put $k = q - 2$.

In the positive branch with $a > 0$, congruence (10) gives $v = 2^k A + a$, hence

$$e(v) = (-1)^k e(A) + e(a).$$

The assumed bound $v < v_q$ leaves at most $2m - 1$ digits in A when q is even and at most $2m$ when q is odd. In both cases $(-1)^k e(A) \leq m$, so

$$e(v) \leq m + e(a) \leq m + a < m + 4a,$$

a contradiction. If $a = 0$, then $2^{q-2} \mid v$. Since $v \in V$, its trailing-zero count is at least s_q; deleting those even-many zeroes leaves an integer with alternating digit sum at least m, hence no smaller than U_m. Therefore $v \geq v_q$.

In the negative branch, $a = 0$ is impossible: after deleting the forced trailing zeroes, the core has at most $2m - 1$ digits and starts in a positive-sign position, so its alternating digit sum cannot be $-m$. If $a > 0$, then

$$v = 2^k A + (2^k - a), \qquad e(2^k - a) = e(2^k - 1) - e(a - 1).$$

Here $e(2^k - 1)$ is zero for even q and one for odd q. Combining this with the same bound on A gives

$$e(v) \geq -m - a + 2 > -m - 4a,$$

again a contradiction. This proves the first assertion.

Now assume $2^{q+2} \mid H(v)$. For even q, again suppose $v < v_q$, but split off the low q bits in (10). In the positive branch, the low block is $2^{q-2} + a$, contributing $1 + e(a)$, while the high block has at most $2m - 3$ digits and contributes at most $m - 1$. Thus $a > 0$ again gives $e(v) < m + 4a$. When $a = 0$, equality forces the high block to be U_{m-1}, and

$$2^q U_{m-1} + 2^{q-2} = 2^{q-2} U_m = v_q.$$

In the negative branch, the low block is $3 \cdot 2^{q-2} - a$. Its contribution is zero for $a = 0$ and at least $-a$ for $a > 0$, while the high block contributes at least $-(m-2)$. Neither case reaches $-m - 4a$. Hence the even-q minimum is still v_q.

Finally let q be odd and suppose $v \leq 4v_q$. Then v has at most $2m + q$ digits, so

$$|e(v)| \leq m + (q+1)/2.$$

Together with (11), this implies $a < 2^{q-2}$. In the positive branch, the low q bits are $2^{q-2} + a$ and contribute $-1 + e(a)$, while the high block contributes at most m; hence

$$e(v) \le m - 1 + e(a) < m + 4a.$$

In the negative branch, $a = 0$ would force $\nu_2(v) = q - 2$, which is odd and contradicts $v \in V$. For $a > 0$,

$$e(3 \cdot 2^{q-2} - a) = 2 - e(a-1) \ge 3 - a,$$

so $e(v) \ge -m + 3 - a > -m - 4a$. Thus $v > 4v_q$. Since H is strictly increasing and $H(4v_q) = 4H(v_q) + m$, we obtain $H(v)/4 > H(v_q)$. This proves Lemma 4. □

We now spell out the four member-side cases needed for the lower bound. Put $x = 2^q n$ and suppose $d(4x) \ne d(x)$. By (5), exactly one side belongs to $H(V)$.

- If $x = H(v)$ and q is odd, Lemma 4 gives $x \ge H(v_q) = 2^q M_q$.
- If $x = H(v)$ and q is even, it gives $x \ge H(v_q) = 2^{q+2} M_q > 2^q M_q$.
- If $4x = H(v)$ and q is even, the strong-divisibility part gives $x = H(v)/4 \ge H(v_q)/4 = 2^q M_q$.
- If $4x = H(v)$ and q is odd, it gives $x = H(v)/4 > H(v_q) = 2^q M_q$.

Consequently $n \ge M_q$ in every case. The witnesses in Section 3 give the reverse inequality, proving Theorem 1. Explicitly,

$$M_q = \begin{cases} (4^{2^q-1} - 1)/9, & q \text{ even}, \\ (2 \cdot 4^{2^q} - 5)/9, & q \text{ odd}. \end{cases}$$

6 Arbitrary distances and multipliers

The values $M_q, M_{q+2}, M_{q+4}, \ldots$ are strictly increasing. For $n < M_q$, no adjacent pair in

$$d(2^q n), d(2^{q+2} n), \ldots, d(2^{q+2r} n)$$

has separated. At $n = M_q$, the first pair separates while all later adjacent pairs remain equal. Therefore

$$\min\{n \ge 1 : d(2^{q+2r} n) \ne d(2^q n)\} = M_q$$

for every $r \ge 1$. Substituting $b = 2^j u$ proves (2), and $u \mid M_q$ gives the exact witness $n = M_q/u$. For even q, $M_q = (2^{E_q} - 1)/9$. If $L = \mathrm{ord}_{9u}(2)$, then

$$u \mid M_q \iff L \mid 2(2^q - 1).$$

This has infinitely many even solutions q exactly when the 2-adic valuation of L is at most one. Indeed, write $L = 2^a s$ with s odd. Necessity follows because $\nu_2(2(2^q - 1)) = 1$. Conversely, if $a \leq 1$, choose q divisible by $\operatorname{lcm}(2, \operatorname{ord}_s(2))$; then q is even and $s \mid 2^q - 1$, so $L \mid 2(2^q - 1)$. When $s = 1$, any even q works. Taking arbitrary positive multiples of the chosen q gives infinitely many solutions and proves (3).

7 Scope

Theorems 1 and 2 solve an infinite ratio-four subfamily of the general first-separation question. They do not determine the exact answer when $u \nmid M_q$, nor do they solve the problem for two arbitrary multipliers. In the nondivisibility case, the natural finite-state descriptions retain all residue classes modulo u, and no parameter-independent second-minimum theorem is presently known.

References

[1] G. Allouche, J.-P. Allouche, and J. Shallit, *Kolam indiens, dessins sur le sable aux îles Vanuatu, courbe de Sierpiński et morphismes de monoïdes*, Annales de l'Institut Fourier **56** (2006), 2115–2130. doi:10.5802/aif.2235.

[2] N. J. A. Sloane, *The On-Line Encyclopedia of Integer Sequences*, sequence A015565. OEIS A015565.

[3] J. Shallit, *Open Problems in Automata Theory and Formal Languages*, Open Problem 7. `https://cs.uwaterloo.ca/~shallit/Talks/open10r.pdf`.

AI-assisted research and writing disclosure. Carptopus is the author responsible for the mathematical claims and the public version of this manuscript. OpenAI Codex was used as an AI-assisted tool for literature organization, proof development and stress testing, exact verification, adversarial auditing, and manuscript preparation.